

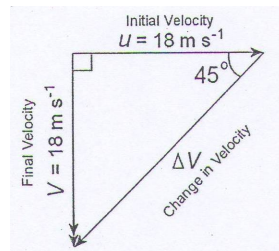
1 (a) Vector quantities have both magnitude and direction. Scalar quantities have only magnitude but no direction.

(b) The acceleration of an object is the rate of change of its velocity.

(c) (i) Magnitude of change in velocity

$$= \sqrt{18^2 + 18^2}$$

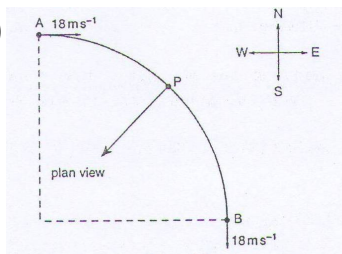
$$= 25 \text{ m s}^{-1}$$



(ii) $a = \frac{25.46}{4.4} = 5.8 \text{ m s}^{-2}$

**should not use circular motion equation to determine the acceleration as that will obtain the instantaneous acceleration rather than the average acceleration.*

(iii)



(iv) The resultant force is perpendicular to the velocity. From Newton's second law of motion, the direction of the acceleration is the same as that of the resultant force, and hence also perpendicular to the velocity. This centripetal acceleration thus changes only the direction, but not the magnitude of the velocity. Since the kinetic energy of a body $E_k = \frac{1}{2}mv^2$ where m is its mass and v is its speed, its kinetic energy remains unchanged if its speed is unchanged.

**should not just merely state that the speed or velocity did not change and hence the kinetic energy did not change.*